17/03/2024. T-7. (KNN)

Classification

Action Scenes	Romance Scenes	Happy Ending	Genre
15	2	0	A
1	12	1	R
18	1	0	A
2	14	1	R
14	3	1	A

Test DataSet

Test ID	Action Scene	Romance Scene	Happy Ending	Actual Genre
T_1	10	5	1	A.

Sol:

(10, 5, 1) → Testing Features.

* Calculate distance from test data to each input data point

(10, 5, 1) — (15, 2, 0)
(1, 12, 1)
(18, 1, 0)
(2, 4, 1)
(14, 3, 1).

$$\sqrt{(x_{test}-x_1)^2+(x_{test}-x_2)^2+(x_{test}-x_3)^2}$$

$d_1 = \sqrt{(10-15)^2+(5-2)^2+(1-0)^2} = d_1 = 5.92$ — A

$d_2 = \sqrt{(10-1)^2+(5-12)^2+(1-1)^2} = d_2 = 11.90$ — R

$d_3 = \sqrt{(10-18)^2+(5-1)^2+(1-0)^2}$ $d_3 = 9.0$ — A

$d_4 = \sqrt{(10-2)^2+(5-4)^2+(1-1)^2}$ $d_4 = 12.04$ — R.

$d_5 = \sqrt{(10-14)^2+(5-3)^2+(1-1)^2}$ $d_5 = 4.47$ — A

K=3

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Weight Sense $= \frac{1}{d}$

$w = 1/d$

- A $(w_1 = 0.19)$
- R $(w_2 = 0.08)$
- A $(w_3 = 0.11)$
- R $(w_4 = 0.08)$
- A $(w_5 = 0.22)$

weights of A $= (w_1 + w_3 + w_5)$
$= 0.52$

weights of R $= (w_2 + w_4)$
$= 0.16$

Suppose
KNN $(K = 4)$

$d_1 = Y$
$d_2 = Y$
$d_3 = N$
$d_4 = N$

$w_1 = \frac{1}{2}$

Regression

Age (x_1)	Previous value (x_2) (in Lakh)	Car Price (Y) (in Lakh)
2	3	2
3	4	3
5	1	0.5

Test : $(4, 2, ?)$ → Predicting Y - Goal.

Distance. $(4,2)$ — $(2,3)$
— $(3,4)$
— $(5,1)$

$d_1 = \sqrt{(4-2)^2 + (2-3)^2} = \sqrt{5} = 2.24$
$d_2 = \sqrt{(4-3)^2 + (2-4)^2} = 2.24$
$d_3 = \sqrt{(4-5)^2 + (2-1)^2} = 1.41$ → shortest dist

0.5

Pf $\boxed{k=2}$

$$\boxed{\hat{y} = \frac{1}{k} \sum_{i=1}^{k} y_i}$$

$$= \frac{1+0.5}{2}$$

$$= 1.75$$

$$\hat{y} = 1.75.$$